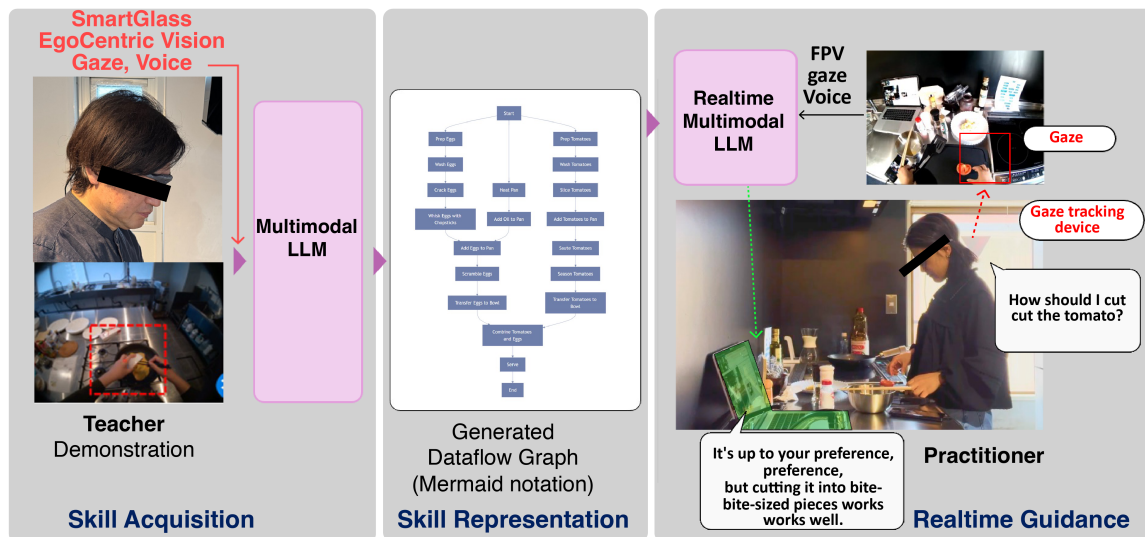


# MermaidLLM: Dataflow Diagrams for Explainable Skill Formalization and Real-time Support with Multimodal LLMs

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**Figure 1:** MermaidLLM uses MLLMs to comprehend expert skills from egocentric vision/gaze, formalizing them as dataflow diagrams (Mermaid). It then acts as a real-world agent, comparing the learner’s context with the diagram and providing verbal guidance.

## Abstract

Traditional skill acquisition methods require laborious manual authoring, hindering scalability. In contrast, advances in Multimodal Large Language Models (MLLMs), combined with wearable sensors that capture egocentric vision, gaze, and voice data, create new opportunities for AI to understand and transfer human skills. We propose “MermaidLLM”, where MLLMs process multimodal input to automatically generate explainable dataflow diagrams of the task description in Mermaid notation. This intermediate task representation improves the shared human-AI understanding, facilitates verification of the comprehension of the MLLM, and simplifies analytical comparisons. During skill transfer, MermaidLLM operates as a real-time agent, using these diagrams to offer trainees immediate, context-sensitive guidance through wearable devices. We evaluated

this approach through user studies, demonstrating the effectiveness of Mermaid diagrams for skill understanding and the potential of the real-time agent.

## CCS Concepts

• **Human-centered computing** → Sound-based input / output; Interface design prototyping; Mobile devices.

## Keywords

LLM, Multimodal-LLM, multimodal interface, skill transfer, dataflow, 1st-person vision, gaze, artificial intelligence, neural networks

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## 1 Introduction

Transferring complex human skills, from scientific experiments to cooking, is vital but challenging. Traditional methods typically require manual formalization, such as text manuals or laboratory notebooks, making the process time-consuming, inflexible, and challenging to scale. Although recent multimodal LLMs (MLLMs) effectively process egocentric video to interpret actions [13], their internal processes remain opaque (“black-box”), complicating verification and potentially resulting in inaccurate or misleading guidance similar to linguistic hallucinations. In this paper, we introduce *MermaidLLM* (Figure 1), a system in which an MLLM analyzes egocentric vision, eye-tracking, and voice annotations to generate a data flow diagram using the Mermaid notation [1]. This approach offers several advantages.

- **Shared Understanding:** dataflow diagrams provide a lightweight and human-readable representation, facilitating human-AI collaboration.
- **Verifiability:** The explicit diagram allows experts to verify the comprehension of MLLM.
- **Analysis:** Differences between multiple skill demonstrations can be analyzed by comparing or merging diagrams.
- **Efficiency:** Dataflow diagrams are more compact than video, suitable for formalizing diverse actions. LLMs often have preexisting knowledge of Mermaid syntax.

During skill transfer, *MermaidLLM* acts as a real-time agent. A trainee wearing smart glasses receives guidance based on the expert-derived Mermaid diagram. The system compares the trainee’s actions (observed via sensors) with the diagram, providing context-aware verbal instructions and feedback.

## 2 Related Work

Human activity recognition and assistant systems such as PrISM [2] use Markov models in IMU data, but require manual efforts to represent the task. Egocentric vision [7] and gaze [5] offer a richer input to recognize interaction and tasks of objects. GazeLLM [13] specifically uses gaze-informed video MLLMs for activity description, which *MermaidLLM* builds on by adding a structured representation and a real-time interactive agent component.

Using LLMs to generate task representations like dataflow diagrams [9], clinical trees [10], analysis plans [15], or robot plans [11, 14] is an emerging area. *MermaidLLM* uniquely applies this to human skill transfer using multimodal input from wearables, focusing on generating a verifiable intermediate representation and using it for real-time human guidance. Wearable devices such as Meta Aria [6] and Pupil Labs Neon [12] provide the necessary sensing capabilities.

## 3 MermaidLLM

*MermaidLLM* operates in two main stages:

- **Skill formalization:** An expert performs a task while wearing a device that captures egocentric video, gaze, and optionally voice narration. An MLLM processes this data to generate a Mermaid dataflow diagram representing the task’s procedural structure.
- **Real-time Assistance:** A novice wears smart glasses during task practice. The system uses the pregenerated Mermaid diagram

as a reference, compares it with the novice’s real-time actions (via video/gaze), and provides context-aware verbal guidance through interaction with an MLLM.

### 3.1 System Configuration

We use Google Gemini Pro 2.5 as an MLLM, and use the EgoExo4D dataset [8] and our custom dataset for skill formalization. For real-time assistance, we use Pupil Core Neon glasses [12], OpenCV-Python for gaze overlay, Pupil Labs Realtime API for data streaming, and Google’s Gemini Live API via WebSocket for low-latency multimodal interaction. The MLLM prompts are defined for role definition, awareness of real-time video/gaze input, and the task-specific Mermaid diagram. Users can ask for guidance through voice during task practice.

## 4 Pilot User Study of Real-time Interaction

As a pilot user study, six participants (with various cooking frequencies) were recruited and performed the following task:

- (1) Participants were briefed, consented to, and given a printed MLLM-generated Mermaid diagram for a “tomato and egg stir fry” task.
- (2) Wore Pupil Core Neon eye tracking glasses.
- (3) Performed the cooking task in a real kitchen, interacting with the real-time *MermaidLLM* agent via voice queries (for example, “What is next?”, “How do I season this?”).
- (4) Completed the System Usability Scale (SUS) [4] questionnaire post task.
- (5) Participated in a semi-structured interview about their experience.

### 4.1 Results

The mean SUS score was 68.3 ( $std = 13.5$ ), indicating acceptable usability (“OK” to “Good” range [3]).

The system demonstrated an effective contextual understanding of the user’s activity by combining egocentric vision, gaze tracking, object recognition, and voice input to deliver pertinent verbal guidance. (e.g., identification of the seasoning bottle near the gaze point when asked how to season). The required granularity of the instruction varied significantly based on the user’s experience (newer versions needed more detail). The post-interviews also revealed:

- **Usefulness:** Many found real-time voice guidance useful, especially hands-free. Some found the interaction pleasant, reducing isolation.
- **Agent Responsiveness:** Quick responses valued. The agent sometimes gave the next instruction prematurely, indicating imperfect state tracking.
- **Positive Support:** Strong desire for positive feedback/praise from the agent especially when everything is going right.
- **Future Applications:** Many participants suggested that this system can be used for childcare, DIY, and end-to-end technical support, especially for novices.

## 5 Conclusion

We present a method to let MLLMs leverage multimodal inputs (egocentric vision, gaze, voice) from wearables and comprehend

human skills and represent them as dataflow notations. This representation facilitates the shared understanding and verification of human-AI relationships. The pilot user study demonstrated the potential of a real-time agent using this representation to provide contextual guidance for practitioners.

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## References

- [1] [n. d.]. Mermaid: Diagramming and charting tool. <https://mermaid.js.org/>
- [2] Riku Arakawa, Hiromu Yakura, and Mayank Goel. 2024. PrISM-Observer: Intervention Agent to Help Users Perform Everyday Procedures Sensed using a Smartwatch. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 15, 16 pages. doi:10.1145/3654777.3676350
- [3] Aaron Bangor, Philip Kortum, and James Miller. 2009. Determining what individual SUS scores mean: adding an adjective rating scale. *J. Usability Studies* 4, 3 (May 2009), 114–123.
- [4] John Brooke et al. 1996. SUS-A quick and dirty usability scale. *Usability evaluation in industry* 189, 194 (1996), 4–7.
- [5] Haodong Chen, Niloofar Zendejdel, Ming C. Leu, and Zhaozheng Yin. 2025. A gaze-driven manufacturing assembly assistant system with integrated step recognition, repetition analysis, and real-time feedback. *Engineering Applications of Artificial Intelligence* 144 (2025), 110076. doi:10.1016/j.engappai.2025.110076
- [6] Jakob Engel et al. 2023. Project Aria: A New Tool for Egocentric Multi-Modal AI Research. arXiv:2308.13561 [cs.HC] <https://arxiv.org/abs/2308.13561>
- [7] Alireza Fathi, Xiaofeng Ren, and James M. Rehg. 2011. Learning to recognize objects in egocentric activities. In *CVPR 2011*. 3281–3288. doi:10.1109/CVPR.2011.5995444
- [8] Kristen Grauman et al. 2024. Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. arXiv:2311.18259 [cs.CV] <https://arxiv.org/abs/2311.18259>
- [9] Guntur Budi Herwanto. 2024. Automating Data Flow Diagram Generation from User Stories Using Large Language Models. In *CEUR*. Winterthur, Switzerland. <https://hal.science/hal-04525925>
- [10] Binbin Li, Tianxin Meng, Xiaoming Shi, Jie Zhai, and Tong Ruan. 2023. MedDM: LLM-executable clinical guidance tree for clinical decision-making. *ArXiv abs/2312.02441* (2023). <https://api.semanticscholar.org/CorpusID:265658947>
- [11] David Paulius, Alejandro Agostini, Benedict Quarthey, and George Konidaris. 2025. Bootstrapping Object-Level Planning with Large Language Models. In *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*.
- [12] pupil labs. [n. d.]. NEON: Eye tracking for research and beyond. <https://pupil-labs.com/products/neon>
- [13] Jun Rekimoto. 2025. GazeLLM: Multimodal LLMs Incorporating Human Visual Attention. In *Proceedings of Augmented Humans 2025* (Abu Dhabi, UAE) (AHs '25).
- [14] Ryunosuke Takebayashi, Vitor Hideyo Isume, Takuya Kiyokawa, Weiwei Wan, and Kensuke Harada. 2025. Cooking Task Planning using LLM and Verified by Graph Network. arXiv:2503.21564 [cs.RO] <https://arxiv.org/abs/2503.21564>
- [15] Yuheng Zhao, Junjie Wang, Linbin Xiang, Xiaowen Zhang, Zifei Guo, Cagatay Turkay, Yu Zhang, and Siming Chen. 2024. LightVA: Lightweight Visual Analytics with LLM Agent-Based Task Planning and Execution. *IEEE Transactions on Visualization and Computer Graphics* (2024), 1–13. doi:10.1109/TVCG.2024.3496112